

Geolocalización en Android y Permisos

1. Permisos en Android

Para usar ubicación necesitas declararlos en el AndroidManifest.xml:

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
```

ACCESS_FINE_LOCATION → ubicación precisa (GPS)

ACCESS_COARSE_LOCATION → ubicación aproximada (WiFi/red)

2. Permisos en tiempo de ejecución

```
if (checkSelfPermission(android.Manifest.permission.ACCESS_FINE_LOCATION)
    != PackageManager.PERMISSION_GRANTED) {

    requestPermissions(
        arrayOf(android.Manifest.permission.ACCESS_FINE_LOCATION),
        100
    )
}

override fun onRequestPermissionsResult(
    requestCode: Int,
    permissions: Array<String>,
    grantResults: IntArray
) {
    if (requestCode == 100) {
        if (grantResults.isNotEmpty() &&
            grantResults[0] == PackageManager.PERMISSION_GRANTED) {

            // Permiso concedido
        } else {
            // Permiso denegado
        }
    }
}
```

3. Obtener ubicación (FusedLocationProvider)

```
implementation("com.google.android.gms:play-services-location:21.0.1")

val fusedLocationClient = LocationServices.getFusedLocationProviderClient(this)

fusedLocationClient.lastLocation
    .addOnSuccessListener { location ->
        if (location != null) {
            val lat = location.latitude
            val lon = location.longitude
        }
    }
```

4. Ubicación en tiempo real

```
val locationRequest = LocationRequest.create().apply {
    interval = 5000
    fastestInterval = 2000
}
```

```

        priority = LocationRequest.PRIORITY_HIGH_ACCURACY
    }

    val locationCallback = object : LocationCallback() {
        override fun onLocationResult(result: LocationResult) {
            for (location in result.locations) {
                val lat = location.latitude
                val lon = location.longitude
            }
        }
    }

    fusedLocationClient.requestLocationUpdates(
        locationRequest,
        locationCallback,
        Looper.getMainLooper()
    )

```

5. Errores comunes

- No pedir permisos
- Usuario los rechaza
- GPS desactivado
- No comprobar permisos antes de usar la API

6. Permiso en segundo plano

```
<uses-permission android:name="android.permission.ACCESS_BACKGROUND_LOCATION"/>
```

Resumen

1. Declarar permisos
2. Pedir permisos en runtime
3. Usar FusedLocationProviderClient
4. Comprobar que la ubicación no sea null